

COMBINATIONAL LOGIC DESIGN

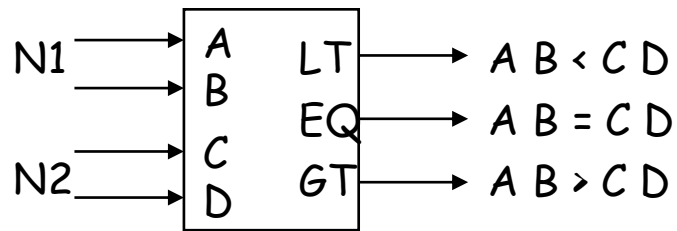
Module.2

CDA 3103C

Spring 2026

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Design example: two-bit comparator



block diagram
and
truth table

Note:

A	B	A<B	A=B	A>B
0	0	0	1	0
0	1	1	0	0
1	0	0	0	1
1	1	0	1	0

Truth Table of 1-Bit Comparator

A	B	C	D	LT	EQ	GT
0	0	0	0	0	1	0
		0	1	1	0	0
		1	0	1	0	0
		1	1	1	0	0
0	1	0	0	0	0	1
		0	1	0	1	0
		1	0	1	0	0
		1	1	1	0	0
1	0	0	0	0	0	1
		0	1	0	0	1
		1	0	0	1	0
		1	1	1	0	0
1	1	0	0	0	0	1
		0	1	0	0	1
		1	0	0	0	1
		1	1	0	1	0

We'll need a 4-variable Karnaugh map for each of the 3 output functions.

Design example: two-bit comparator

				A
	0	0	0	0
	1	0	0	0
	1	1	0	1
C	1	1	0	0
				B

K-map for LT

				A
	1	0	0	0
	0	1	0	0
	0	0	1	0
C	0	0	0	1
				B

K-map for EQ

				A
	0	1	1	1
	0	0	1	1
	0	0	0	0
C	0	0	1	0
				B

K-map for GT

$$LT = A' B' D + A' C + B' C D$$

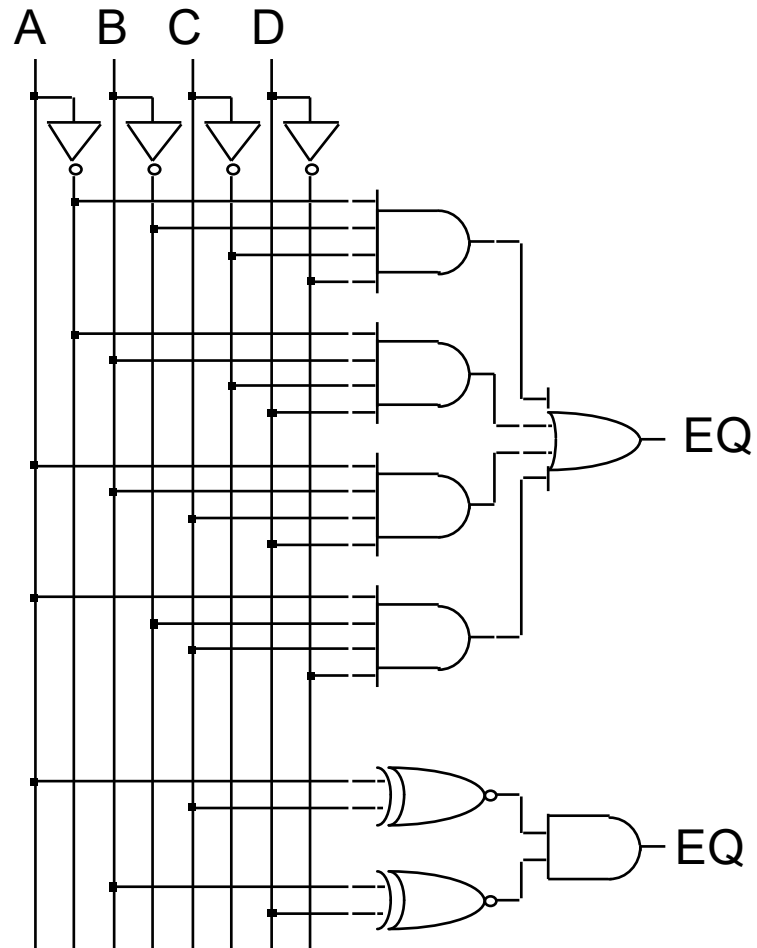
$$EQ = A' B' C' D' + A' B C' D + A B C D + A B' C D'$$

$$= (A \text{ xnor } C) \cdot (B \text{ xnor } D)$$

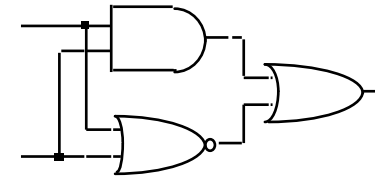
$$GT = B C' D' + A C' + A B D'$$

LT and GT are similar (flip A/C and B/D)

Design example: two-bit comparator

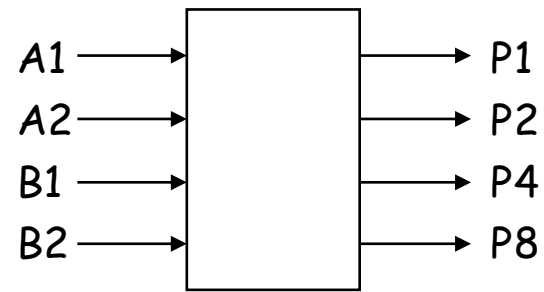


two alternative
implementations of EQ
with and without XNOR



XNOR is implemented with
at least 3 simple gates

Design example: 2x2-bit multiplier

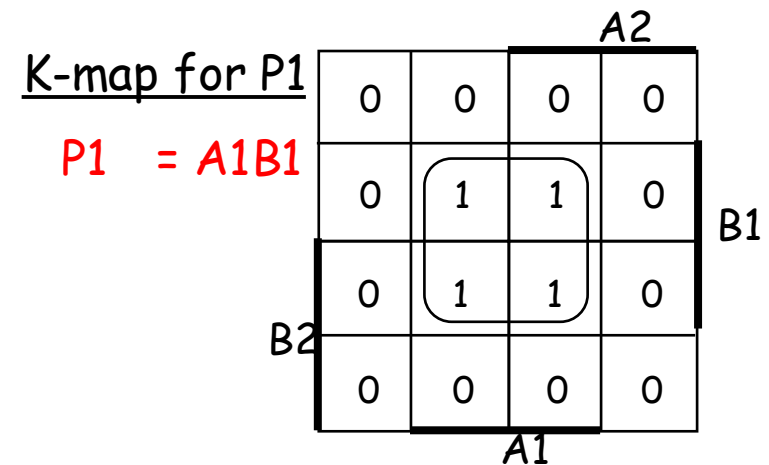
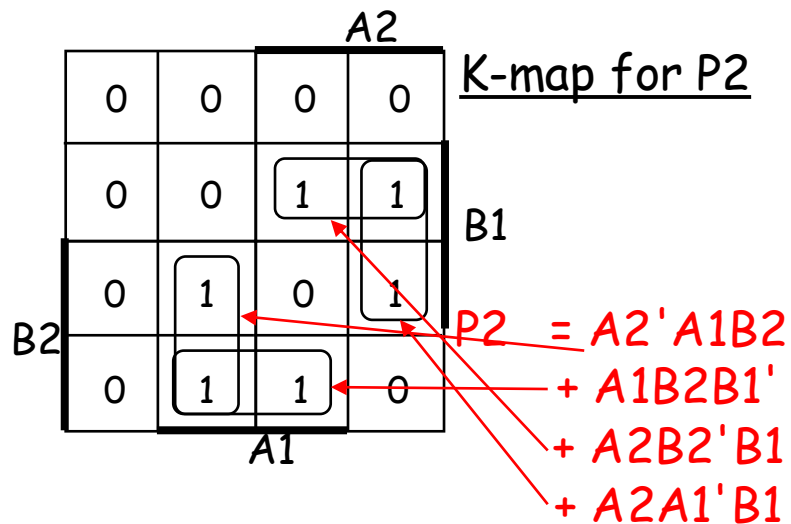
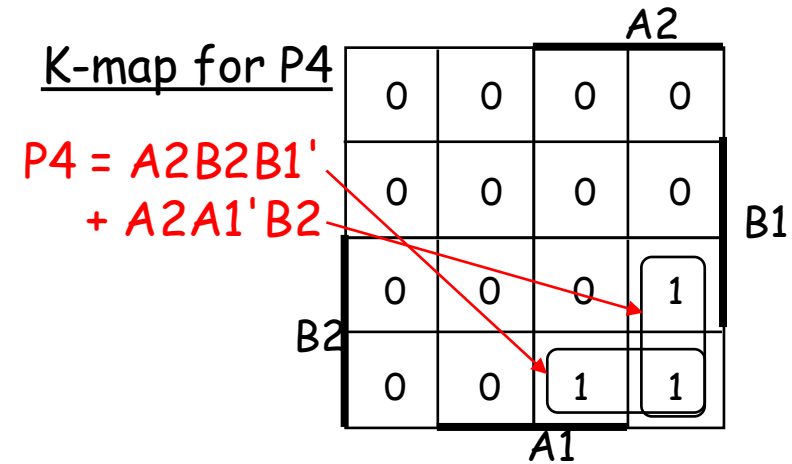
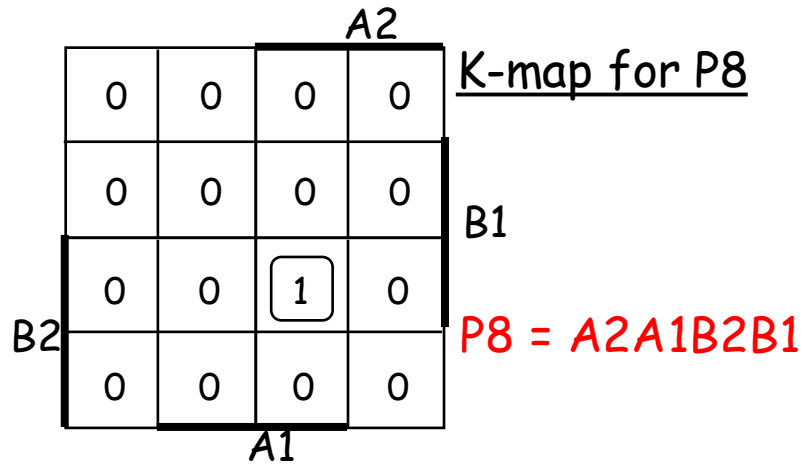


block diagram
and
truth table

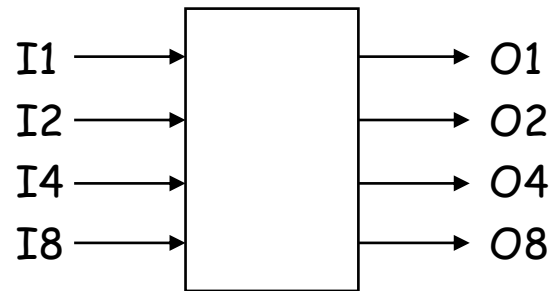
A2	A1	B2	B1	P8	P4	P2	P1
0	0	0	0	0	0	0	0
		0	1	0	0	0	0
		1	0	0	0	0	0
		1	1	0	0	0	0
0	1	0	0	0	0	0	0
		0	1	0	0	0	1
		1	0	0	0	1	0
		1	1	0	0	1	1
1	0	0	0	0	0	0	0
		0	1	0	0	1	0
		1	0	0	1	0	0
		1	1	0	1	1	0
1	1	0	0	0	0	0	0
		0	1	0	0	1	1
		1	0	0	1	1	0
		1	1	1	0	0	1

4-variable K-map
for each of the 4
output functions

Design example: 2x2-bit multiplier (cont'd)



Design example: BCD increment by 1



block diagram
and
truth table

I8	I4	I2	I1	O8	O4	O2	O1
0	0	0	0	0	0	0	1
0	0	0	1	0	0	1	0
0	0	1	0	0	0	1	1
0	0	1	1	0	1	0	0
0	1	0	0	0	1	0	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	1	1
0	1	1	1	1	0	0	0
1	0	0	0	1	0	0	1
1	0	0	1	0	0	0	0
1	0	1	0	X	X	X	X
1	0	1	1	X	X	X	X
1	1	0	0	X	X	X	X
1	1	0	1	X	X	X	X
1	1	1	0	X	X	X	X
1	1	1	1	X	X	X	X

4-variable K-map for each of
the 4 output functions

Design example: BCD increment by 1 (cont'd)

			I8	
	0	0	X	1
	0	0	X	0
	0	1	X	X
I2	0	0	X	X
			I4	
				<u>O8</u>
				I1

$$O8 = I4 I2 I1 + I8 I1'$$

$$O4 = I4 I2' + I4 I1' + I4' I2 I1$$

$$O2 = I8' I2' I1 + I2 I1'$$

$$O1 = I1'$$

			I8	
	0	1	X	0
	0	1	X	0
	1	0	X	X
I2	0	1	X	X
			I4	
				<u>O4</u>
				I1

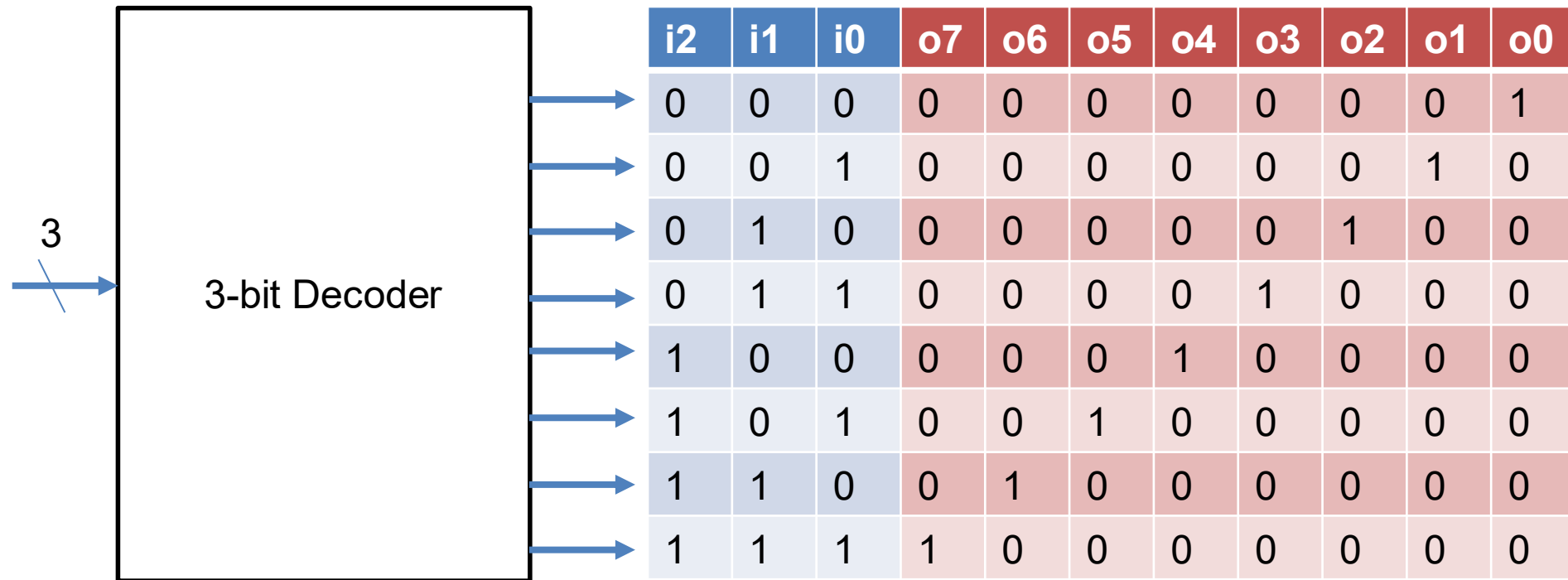
			I8	
	0	0	X	0
	1	1	X	0
	0	0	X	X
I2	1	1	X	X
			I4	
				<u>O2</u>
				I1

			I8	
	1	1	X	1
	0	0	X	0
	0	0	X	X
I2	1	1	X	X
			I4	
				<u>O1</u>
				I1

Combinational Hardware: Decoders

- A decoder is a logic block that takes in an n -bit input and selects from 2^n outputs.
 - One output is asserted for each possible input combination.
 - Outputs are labeled $\text{Out}_0, \text{Out}_1, \dots, \text{Out}_{2^n - 1}$
 - If the input is k , then Out_k will be true

Combinational Hardware: Decoders

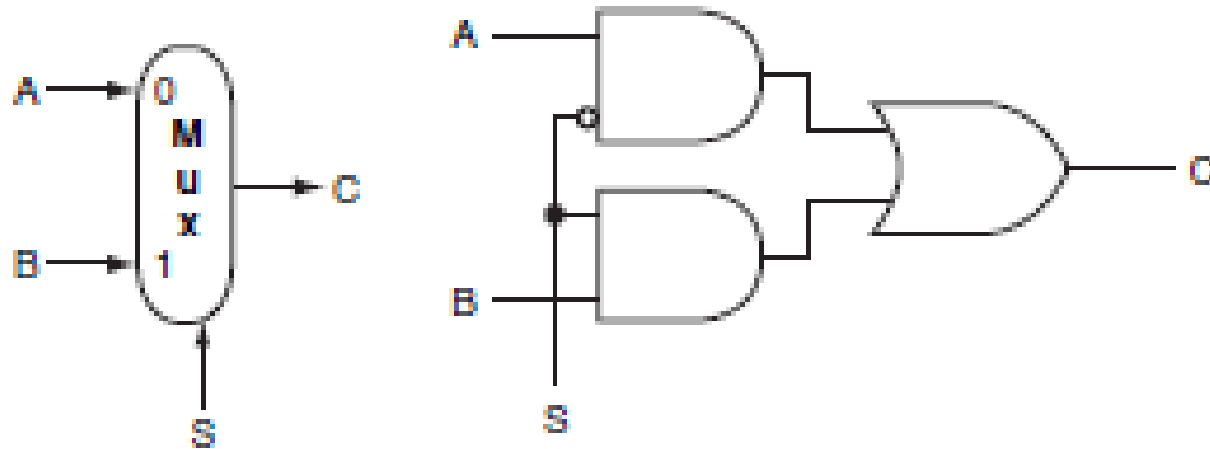


Combinational Hardware: Multiplexors

- A multiplexor is a logic block that takes in n inputs and selects one to be the output.
 - Could also be called a selector
 - The output is one of the inputs, selected by a control value.

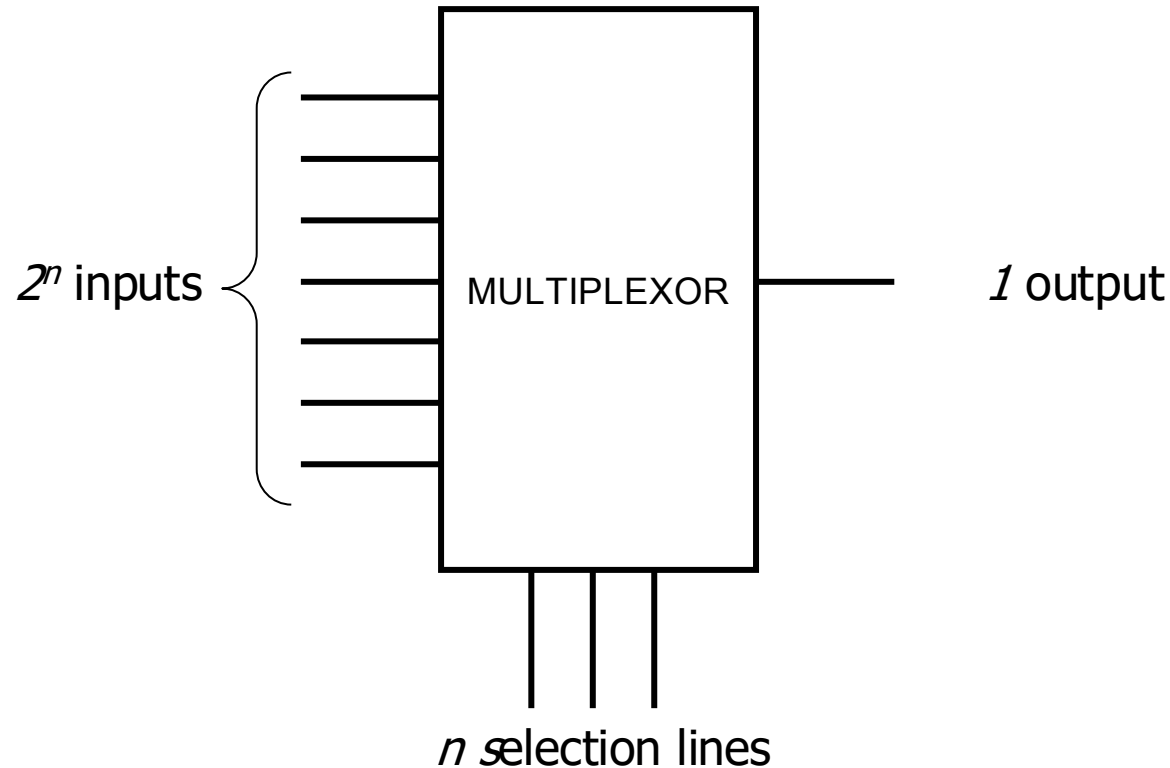
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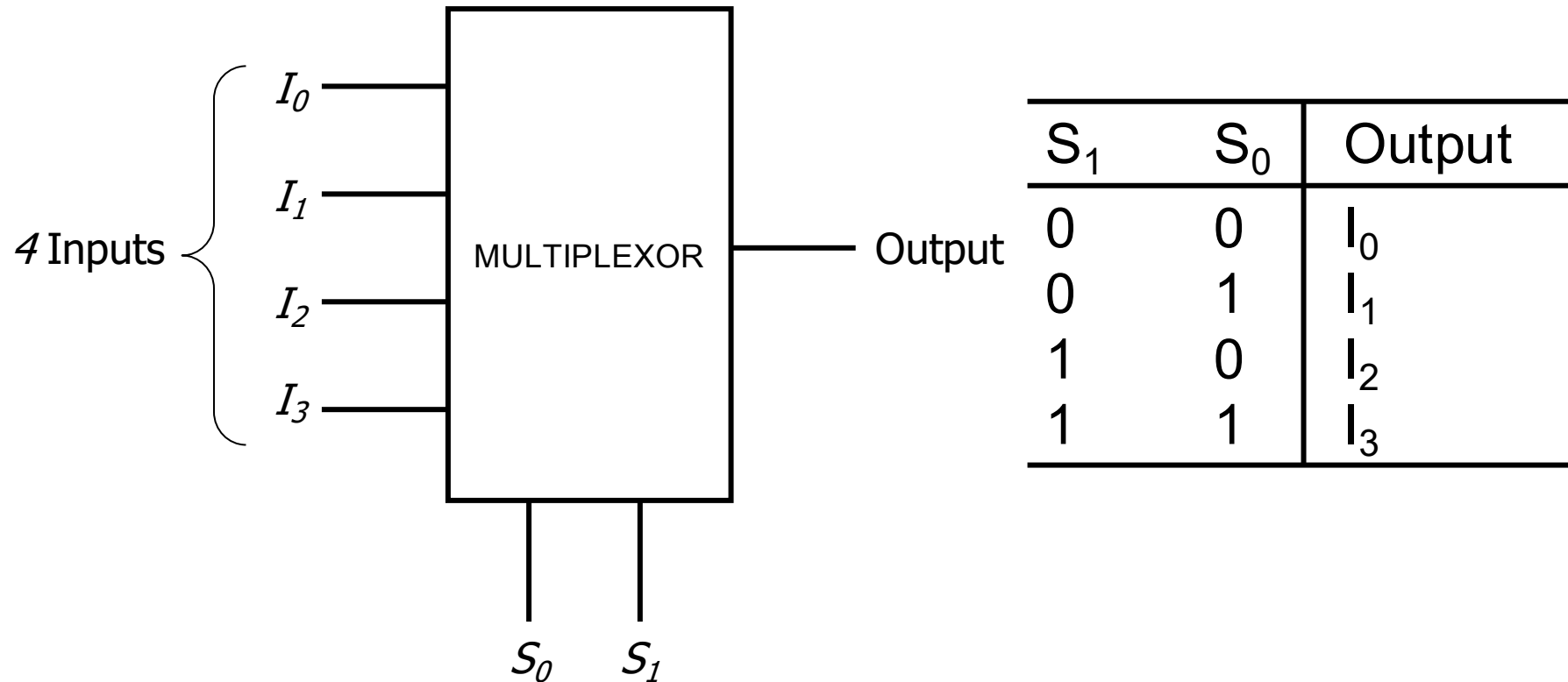


Combinational Hardware: Multiplexors

- Multiplexors can be created with an arbitrary number of data inputs.

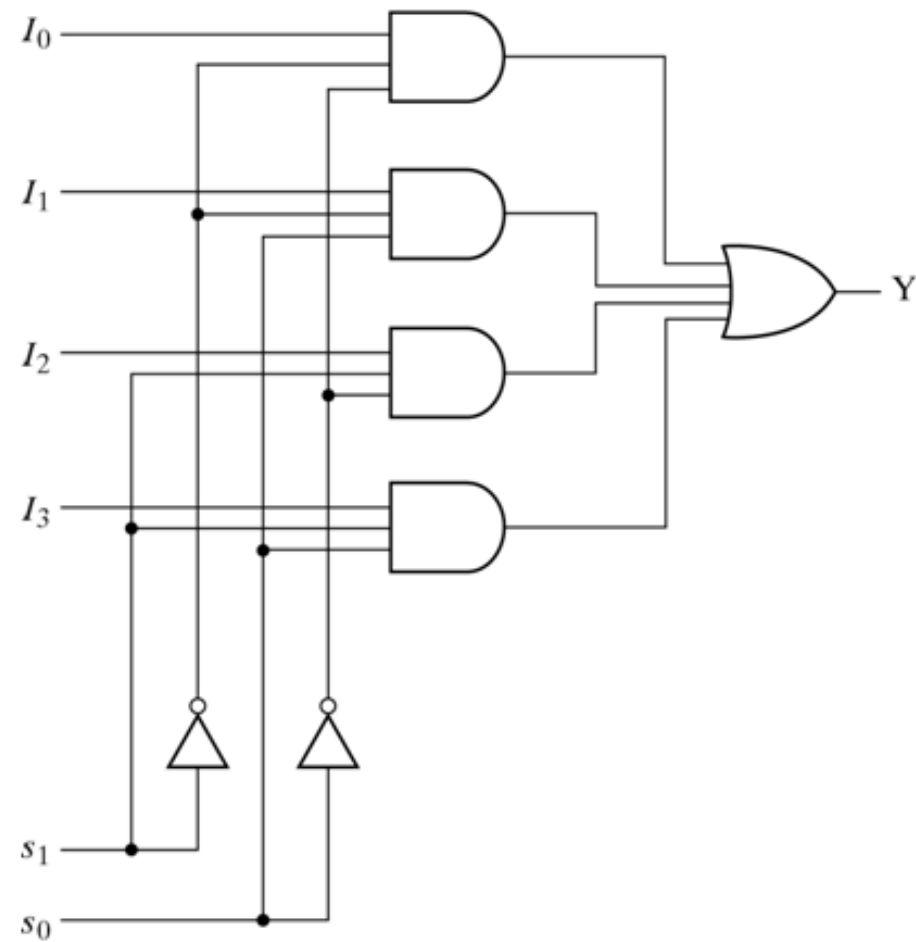


Combinational Hardware: Multiplexors



Combinational Hardware: Multiplexors

- Logic Gates





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